Factors Associated with Instructor Quality Examination by Using Regression Analysis

Shanjie Jiao

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1 Introduction

Universities commonly assess teaching quality through student surveys. However, these evaluations may reflect a range of factors beyond the instructor's teaching ability. This study utilizes student evaluation data from a large American university to explore the relationships among the five dimensions in R(V.4.5.3).

2 Relationships Among the Ratings

Overall, STEM instructors received the lowest average ratings across all five dimensions, especially in easiness, which suggest STEM students have relatively higher difficulty and less satisfied with the teaching quality. In contrast, Pre-prof instructors received the highest average ratings, helpfulness, and easiness, but the lower rate for average interest was only 3.23 which tells that quality ratings may not necessarily correspond to greater interest. Standard deviations were pretty similar across disciplines, but easiness varied a lot more, showing that students had mixed feelings about course difficulty.

As shown in Figure 1, all five rating dimensions were positively correlated. Quality was strongly associated with helpfulness and clarity. In contrast, quality showed weaker correlations with easiness and raterInterest. These relationships are further explored through regression analysis.

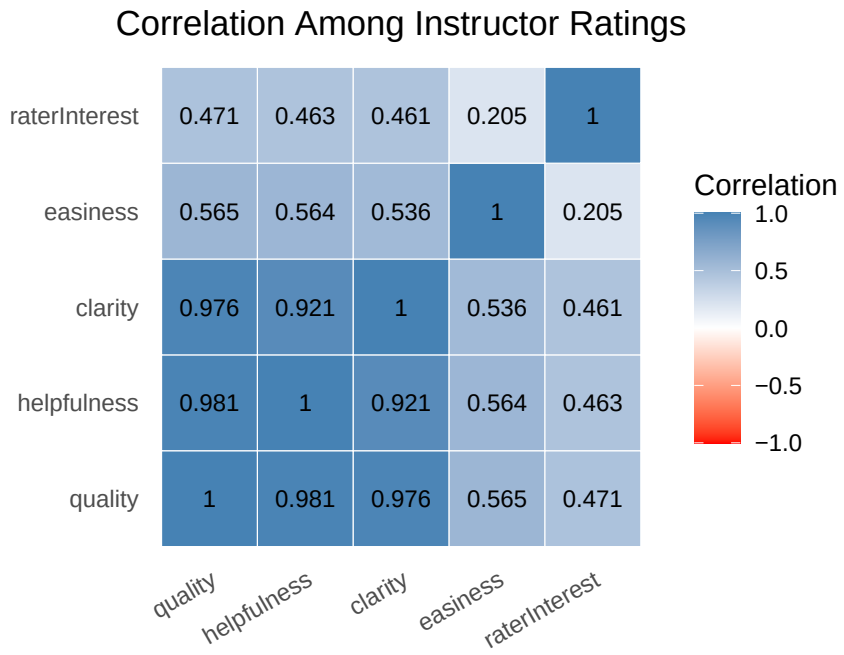

Figure 1

3 Model

We first examined the distributions of numerical variables using histograms and found that quality was slightly left-skewed, while easiness and raterInterest were relatively symmetric. We then used scatterplots to explore their relationships with quality. Both easiness and raterInterest showed positive and approximately linear. Given that quality is a continuous variable, we adopted a normal linear regression, and its assumptions to be assessed through residual diagnostics.

After performing diagnostic plots, the results suggest that the model performs reasonably well, with no major concerns about linearity or normality. Although the Scale-Location plot shows slightly spreaded, but the Breusch-Pagan test found no significant evidence of heterogeneity

with p = 0.133. We also identified 14 potentially influential observations by Cook's distance, while sensitivity analysis showed stable results after being excluded. Meanwhile, the residuals showed no clear pattern across observations, with VIF 1. So, multicollinearity and independent are satisfied. Based on these findings, our final model is below.

$$
\begin{aligned}
Quality_i \mid \mu_i, \sigma &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \beta_0 + \beta_1 \cdot \text{gender}_i + \beta_2 \cdot \text{numYears}_i \\
&\quad + \beta_3 \cdot \text{PreProf}_i + \beta_4 \cdot \text{SocSci}_i + \beta_5 \cdot \text{STEM}_i \\
&\quad + \beta_6 \cdot \text{easiness}_i + \beta_7 \cdot \text{raterInterest}_i + \beta_8 \cdot \text{pepper}_i
\end{aligned}
$$

4 Results

As shown in Table 1, easiness, student interest, pepper, and STEM discipline were positively associated with instructor quality after holding other variables ($p < 0.05$). The association with pepper was the most noticeable, as instructors treated as attractive received approximately 0.56 points higher in ratings than those who were not, with other variable fixed. And student interest and easiness also showed substantial associations, with each one-point increase corresponding to 0.54 and 0.51 higher in quality rating, respectively. In contrast, gender and numYears do not show statistically significant associations ($p > 0.05$). The confidence intervals for the significant predictors excluded zero further supports our finding.

Table 1: Multiple Linear Regression Results for Instructor Quality

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.181	0.242	[-0.657, 0.296]	0.457
gender	0.047	0.065	[-0.081, 0.174]	0.472

Table 1: Multiple Linear Regression Results for Instructor Quality

Predictor	Estimate	SE	95% CI	p-value
numYears	0.018	0.010	[-0.002, 0.037]	0.081
disciplinePre-prof	0.097	0.091	[-0.083, 0.276]	0.291
disciplineSocSci	0.019	0.089	[-0.156, 0.193]	0.834
disciplineSTEM	0.295	0.081	[0.135, 0.455]	<0.001
easiness	0.513	0.042	[0.429, 0.596]	<0.001
raterInterest	0.544	0.059	[0.427, 0.661]	<0.001
pepper	0.562	0.099	[0.366, 0.757]	<0.001

5 Quality and Discipline

Although STEM instructors received the lowest average ratings, but after holding other predictors constant the situation changed. The STEM instructors were estimated to receive approximately 0.30 points higher than Humanities instructors. In comparison, neither Pre-Prof nor Social Science showed statistically significant differences from Humanities after adjustment. This reversed result may reflect that the lower quality ratings are more closely related to course easiness and personal interest in STEM courses but not just discipline itself.

A Appendix 1

```
#install.packages("here")
library(tidyverse)
library(dplyr)
library(here)

raw <- read.csv(here("data", "raw", "Rateprof.csv"), header = TRUE)

# cleaning
raw$X <- NULL

raw$gender <- ifelse(raw$gender == "male", 1, 0)
raw$pepper <- ifelse(raw$pepper == "yes", 1, 0)
raw$discipline <- factor(raw$discipline)

write.csv(raw,
    "/Users/shanjiejiao/Desktop/Instructor Ratings Analysis/data/processed/cleaned.csv")
```

```
#install.packages("here")
#install.packages("reshape2")
#install.packages("car")
library(dplyr)
library(here)
library(tidyr)
library(ggplot2)
```

```r
library(knitr)
library(reshape2)
library(car)
library(broom)

data <- read.csv(here("data", "processed", "cleaned.csv"), header = TRUE)
```

```r
library(dplyr)
library(knitr)

# Select the five rating variables
ratings <- c("quality", "helpfulness", "clarity", "easiness", "raterInterest")

# Summary statistics by discipline
discipline_summary <- data %>%
  group_by(discipline) %>%
  summarise(
    N = n(),
    across(all_of(ratings), ~sprintf("%.2f (%.2f)", mean(.x, na.rm = TRUE),
     sd(.x, na.rm = TRUE)))
  )

# Display the table
kable(discipline_summary,
      col.names = c("Discipline", "N", "Quality", "Helpfulness", "Clarity",
      "Easiness", "Rater Interest"),
```

```
      caption = "Instructor Ratings by Discipline (Mean and SD)",
      align = "lcccccc")
```

```
#histogram
variables <- c("quality", "numYears", "easiness", "raterInterest")
dist_data <- data %>%
  pivot_longer(all_of(variables), names_to = "Variable", values_to = "Value")

# let review distributions
ggplot(dist_data, aes(x = Value)) +
  geom_histogram(bins = 20, fill =  "black", color = "white") +
  facet_wrap(~Variable, scales = "free") +
  labs(title = "Distribution of Numerical Variables", x = "Value",
    y = "Frequency") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```

```
scatter_data <- data[, c("quality", "numYears", "easiness", "raterInterest")]



scatter_long <- pivot_longer(scatter_data, -quality,
                             names_to = "Variable", values_to = "Value")

# Plot relationships with quality
ggplot(scatter_long, aes(x = Value, y = quality)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
```

```
  facet_wrap(~Variable, scales = "free_x") +
  labs(title = "Relationships Between Numerical Predictors and Quality",
       x = "Predictor Value", y = "Quality") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```

```
ratings <- data[, c("quality", "helpfulness", "clarity", "easiness", "raterInterest")]


cor_matrix <- cor(ratings)
cor_long <- melt(cor_matrix)


# Plot correlation heatmap
ggplot(cor_long, aes(Var1, Var2, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(value, 3)), size = 3) +
  scale_fill_gradient2(low = "red", mid = "white", high = "steelblue",
                       midpoint = 0, limits = c(-1, 1)) +
  labs(title = "Correlation Among Instructor Ratings",
       x = NULL, y = NULL, fill = "Correlation") +
  coord_fixed() +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text.x = element_text(angle = 30, hjust = 1),
    panel.grid = element_blank()
  )
```

```r
model1 <- lm(quality ~ gender + numYears + discipline + easiness + raterInterest
+ pepper, data = data)
summary(model1)

# Test regression assumptions
par(mfrow = c(2, 2))
plot(model1)
par(mfrow = c(1, 1))

plot(residuals(model1),
     xlab = "Observation Order",
     ylab = "Residuals",
     main = "Residuals vs Observation Order")
abline(h = 0, col = "red")

# Test multicollinearity
vif(model1)

library(lmtest)

# Test heteroscedasticity
bptest(model1)

# Test influential observations
sort(cooks.distance(model1), decreasing = TRUE)[1:10]
cooks <- cooks.distance(model1)
```

```r
which(cooks > 4/nobs(model1))


# Identify potentially influential observations
influential <- cooks > 4 / nobs(model1)


# check and refit model without flagged observations
model_sensitivity <- lm(
  formula(model1),
  data = model.frame(model1)[!influential, ]
)


round(cbind(
  Original = coef(model1),
  Sensitivity = coef(model_sensitivity)
), 3)
```

```r
# Summarize regression results
model_table <- tidy(model1, conf.int = TRUE) %>%
  mutate(
    across(c(estimate, std.error, conf.low, conf.high), ~round(.x, 3)),
    CI = sprintf("[%.3f, %.3f]", conf.low, conf.high),
    p.value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  select(term, estimate, std.error, CI, p.value)


#   table
kable(model_table,
```

```
      col.names = c("Predictor", "Estimate", "SE", "95% CI", "p-value"),
      caption = "Multiple Linear Regression Results for Instructor Quality",
      align = "lcccc")
```

B Appendix 2: AI Use Disclosure

This report used ChatGPT to assist with R coding, data visualization and part of regression diagnostics. Representative prompts included "How should build a table to show the general data?", "How should I check the influetial observation and to assess whether we should remove them" and "Help me to create a heatmap for correlation." AI-generated suggestions were reviewed, modified, and checked against the actual R outputs and the each line of code true meaning before being incorporated into the analysis and report.